

Cyber Security Awareness

Revision Notes + Final Quiz

How to use this PDF

This revision is based on the Cyber Security Awareness training and the study/quiz structure used in our session. Use the notes for revision first, then take the quiz without looking at the answer key.

1. Human Factor & Cyber Crime

- Human behavior matters: Employees can be a security asset or a security liability.
- Human error: Employees may accidentally create security risks or bypass security controls to complete work faster.
- Cyber Crime goals: Data theft, financial theft, overloading systems, blackmail, and hijacking company computers/resources.
- Key principle: Cybersecurity is not only an IT responsibility; employees have an important role in protecting the organization.

2. Phishing & Email Spoofing

- Phishing: A social engineering attack where an attacker impersonates a trusted contact or organization to deceive the victim.
- Common psychological triggers: Fear, urgency, and curiosity.
- Typical targets: Passwords, financial information, identity, or money.
- Email Spoofing: Forging the sender address so an email appears to come from a trusted sender.
- Remember: Phishing is the deception/attack; spoofing is one technique that can make the message appear trustworthy.

3. Malware & Ransomware

- Malware: Malicious software; the general category.
- Ransomware: A type of malware that makes files/data inaccessible and demands payment (ransom) to restore access.
- Relationship: Ransomware is malware, but not all malware is ransomware.

4. Password Attacks & Password Security

Technique	Core idea
Brute Force	Tries many possible password combinations.
Dictionary Attack	Uses a list of common words/passwords or likely combinations.
Keylogger	Records keystrokes typed by the user, potentially capturing passwords.

- Strong password: At least 12 characters, using uppercase, lowercase, numbers, and symbols.
- Unique passwords: Do not reuse the same password across multiple accounts.
- MFA: Strong passwords are not enough by themselves; use an additional factor.

- MFA examples mentioned: Biometrics, security keys, and one-time passcodes via SMS/app.
- Preferred MFA app in the training: Microsoft Authenticator.

5. Insider Threat

- Insider Threat: A security threat originating from someone within the organization.
- Why it matters: An insider may have knowledge of the organization and access to systems or information.
- Important distinction: An insider threat describes the source of the threat; human error describes an unsafe action or mistake. An employee can create a risk without malicious intent.

6. Protection Checklist

Control	What to remember
Strong & Unique Passwords	12+ characters; uppercase/lowercase/numbers/symbols; unique per account.
MFA	Use an additional authentication factor; Microsoft Authenticator is preferred in the training.
Share with Care	Think about what you post, who may see it, and its possible impact; review privacy/security settings.
Keep a Clean Machine	Keep software and devices up to date; automatic updates are recommended.
Backups	Keep secure electronic copies of important information; the training recommends Microsoft OneDrive.
Public Networks	Avoid sensitive logins such as work email or financial services on public Wi-Fi; use a personal/mobile hotspot when appropriate.
Safe Browsing	Update the browser, use strict privacy/security settings, prefer HTTPS, be careful with downloads, and manage cookies/ads/password-saving settings.
Password Manager	Can generate, securely store, and automatically fill long, unique passwords.
Passwordless	The training introduces passwordless sign-in using Microsoft Authenticator for Office 365.

Final Mock Quiz — 20 Questions

Instructions: Answer without using the revision notes. Record your answers as 1-B, 2-B, etc. Answer key is at the end.

1. An attacker sends a fake email pretending to be a trusted person or organization. What is this?

- A) Ransomware
- B) Phishing
- C) Keylogger
- D) Insider Threat

2. What is the main idea behind Email Spoofing?

- A) Encrypting an email
- B) Forging the sender address to make the email appear trusted
- C) Blocking spam
- D) Creating a backup

3. A malicious program records everything a user types on the keyboard.

- A) Dictionary Attack
- B) Brute Force
- C) Keylogger
- D) Phishing

4. A hacker tries a large number of possible password combinations until finding the correct one.

- A) Keylogger
- B) Brute Force
- C) Ransomware
- D) Spoofing

5. A hacker uses a list of common words/passwords to attempt to access an account.

- A) Dictionary Attack
- B) Brute Force
- C) MFA
- D) Malware

6. A malicious program makes a victim's files inaccessible and demands payment.

- A) Phishing
- B) Ransomware
- C) Insider Threat
- D) Spoofing

7. Which statement is correct?

- A) Malware is a type of Ransomware
- B) Ransomware and Malware are unrelated
- C) Ransomware is a type of Malware
- D) Malware only affects smartphones

8. An employee inside the company uses their access and knowledge to create a security risk.

- A) Phishing
- B) Insider Threat
- C) Brute Force
- D) Public Network Attack

9. Why can human behavior create a security risk?

- A) Employees can make mistakes or bypass security controls
- B) Employees always attack the company
- C) Employees cannot access company systems
- D) Only hackers can cause security incidents

10. Which is the strongest recommendation according to the training?

- A) Use one strong password everywhere
- B) Use unique passwords for different accounts
- C) Share passwords with coworkers
- D) Save passwords anywhere convenient

11. According to the training, a strong password should contain:

- A) Only lowercase letters
- B) Only numbers
- C) Uppercase, lowercase, numbers and symbols
- D) Only symbols

12. According to the training, a strong password should be at least:

- A) 6 characters
- B) 8 characters
- C) 10 characters
- D) 12 characters

13. Why is MFA recommended?

- A) Because passwords are no longer needed in every situation
- B) Because a password alone may not be enough protection
- C) Because MFA disables encryption
- D) Because MFA prevents all cyber attacks

14. Which application is specifically preferred in the training for MFA?

- A) Microsoft Authenticator
- B) Microsoft Word
- C) OneDrive
- D) Windows Media Player

15. You are on public Wi-Fi in a café and need to access work email. What is the safest choice according to the training?

- A) Log in normally
- B) Use the public network for financial and work accounts
- C) Avoid accessing sensitive accounts and use a personal/mobile hotspot instead
- D) Disable your phone's security

16. What does Share with Care encourage you to do?

- A) Share everything publicly
- B) Think about what you're posting, who may see it, and its impact
- C) Disable privacy settings
- D) Use the same password everywhere

17. Why should software be kept up to date?

- A) To reduce the risk of Malware and Ransomware
- B) To make passwords shorter
- C) To disable MFA
- D) To make public Wi-Fi safer

18. What is one purpose of a Password Manager?

- A) Generate and securely store long, unique passwords
- B) Remove the need for security
- C) Share passwords publicly
- D) Disable browser security

19. You receive: 'URGENT — Your account will be disabled today. Click here immediately and enter your password.' What is the strongest warning sign?

- A) Urgency + request for credentials
- B) Short email + company logo
- C) Email contains text + subject line
- D) It mentions your account

20. An employee receives a suspicious email, clicks the link, enters their password, and later discovers someone accessed their account. Best description?

- A) The email was a phishing attempt, and the employee's action created a security risk
- B) It was automatically ransomware
- C) It was a Brute Force attack only
- D) It was an Insider Threat because the employee clicked a link

Answer Key

1-B | 2-B | 3-C | 4-B | 5-A | 6-B | 7-C | 8-B | 9-A | 10-B
11-C | 12-D | 13-B | 14-A | 15-C | 16-B | 17-A | 18-A | 19-A | 20-A

Quick Pre-Quiz Checklist

- ☐ Phishing = social engineering / deception.
- ☐ Email Spoofing = forged sender address.
- ☐ Malware = malicious software; Ransomware = a type of malware.
- ☐ Brute Force vs Dictionary vs Keylogger.
- ☐ Insider Threat = threat from within the organization.
- ☐ Strong password = 12+ characters + mixed character types + unique per account.
- ☐ MFA adds another authentication factor; Microsoft Authenticator is highlighted.
- ☐ Think before sharing information; review privacy/security settings.
- ☐ Keep devices/software updated and back up important data.
- ☐ Avoid sensitive logins on public Wi-Fi; use a personal/mobile hotspot when appropriate.
- ☐ Use HTTPS and be careful with downloads.
- ☐ Password Manager can generate, store, and fill unique passwords.

Practice result from this session: 20/20 (100%) on the final mock quiz.